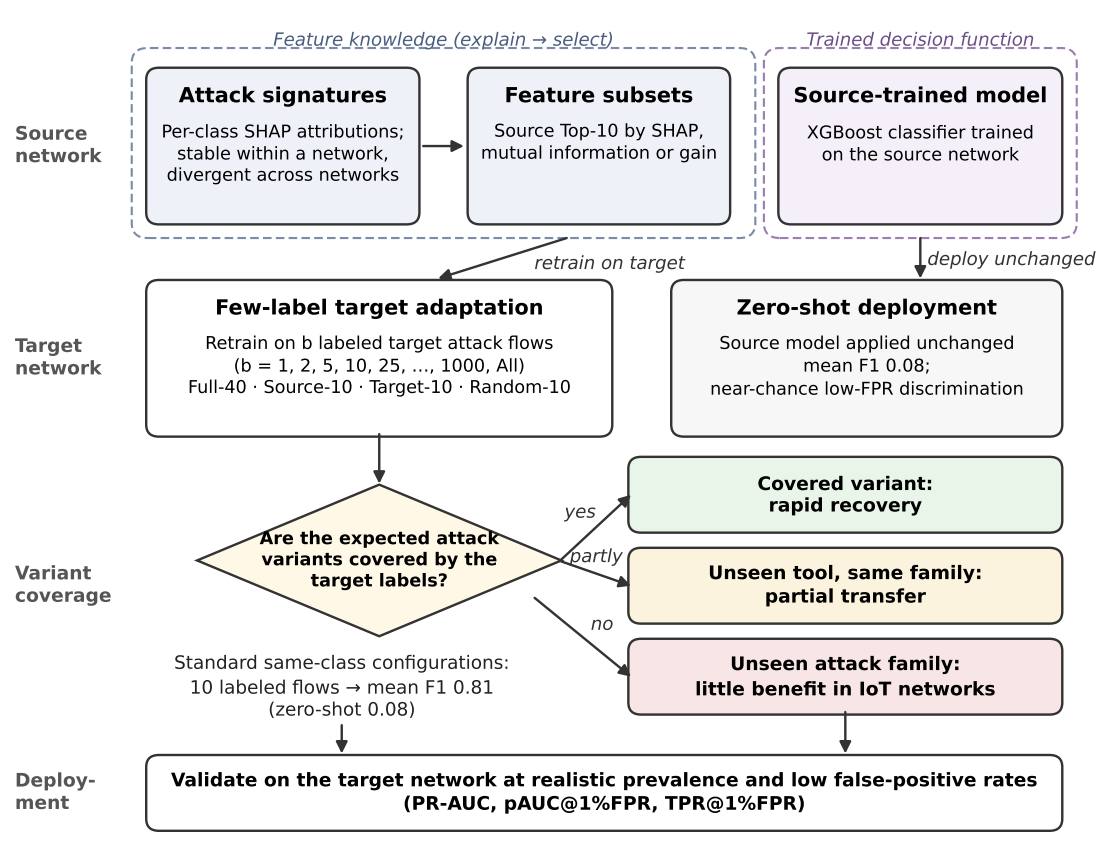
Feature knowledge (explain → select)
Trained decision function
Source network
Attack signatures
Per-class SHAP attributions; stable within a network, divergent across networks
Feature subsets
Source Top-10 by SHAP, mutual information or gain
Source-trained model
XGBoost classifier trained on the source network
retrain on target
deploy unchanged
Target network
Few-label target adaptation
Retrain on b labeled target attack flows
(b = 1, 2, 5, 10, 25, ..., 1000, All)
Full-40 · Source-10 · Target-10 · Random-10
Zero-shot deployment
Source model applied unchanged
mean F1 0.08;
near-chance low-FPR discrimination
Variant coverage
Are the expected attack variants covered by the target labels?
yes
partly
no
Covered variant: rapid recovery
Unseen tool, same family: partial transfer
Unseen attack family: little benefit in IoT networks
Standard same-class configurations:
10 labeled flows → mean F1 0.81
(zero-shot 0.08)
Deploy-ment
Validate on the target network at realistic prevalence and low false-positive rates
(PR-AUC, pAUC@1%FPR, TPR@1%FPR)